Lesson 1: Where Does the Energy Go?

Unit

Energy: Stores and Transfers

Lesson focus

Students learn to describe any change as energy moving between stores. They name the stores, the four ways energy is transferred, and what a system is, then describe the five changes the AQA specification names.

Why this lesson comes first

Every later lesson calculates the energy in one store: kinetic, gravitational or elastic. Students need a reason to calculate it. Here, they learn that energy is never made or used up; it only moves. That is the idea the calculations in Lessons 2–6 test, and the idea that wasted energy and efficiency in Lessons 8–9 depend on.

Specification

AQA GCSE Physics 4.1.1.1 Energy stores and systems.

Learning objectives

By the end of the lesson, students should be able to:

1. Name the energy stores and give an example of each.
2. Name the four ways energy is transferred between stores.
3. Explain what is meant by a system.
4. Describe the changes in energy stores when a ball is thrown upwards, when a moving object hits an obstacle, when a force speeds an object up, when a vehicle slows down and when a kettle boils water.

Key vocabulary

Keyword	Student-friendly definition
Energy	What must be transferred to make something change; measured in joules (J)
Energy store	Where energy is held: kinetic, gravitational potential, elastic potential, thermal, chemical, magnetic, electrostatic, nuclear
Transfer	Energy moving from one store to another
System	The object or group of objects we choose to study
Closed system	A system that no energy enters or leaves; its total energy stays the same
Joule (J)	The unit of energy

The stores

Store	Energy is held because…	Example
Kinetic	an object is moving	A rolling football
Gravitational potential	an object is raised in a gravitational field	A book on a high shelf
Elastic potential	an object is stretched or squashed	A drawn bow, a stretched spring
Thermal	an object is hot (its particles move faster)	A cup of tea
Chemical	of the chemical bonds in a substance	Food, fuel, a battery
Magnetic	magnets attract or repel	Two fridge magnets held apart
Electrostatic	charges attract or repel	A charged balloon near a wall
Nuclear	of the forces holding nuclei together	Uranium fuel

The four ways energy is transferred

Pathway	What happens	Example
Mechanically	A force moves something (sound is carried this way too, by vibrations)	Pushing a trolley
Electrically	A current flows	A battery lighting a lamp
By heating	A hotter object warms a cooler one	A hob heating a pan
By radiation	Light or infrared carries it	The Sun warming your face

Key point:

> *Stores hold energy. Transfers move it. "Light energy" and "sound energy" are not stores: they are ways energy travels from one store to another.*

Starter: What has changed?

Show a photograph or short clip: a skateboarder rolling up a ramp, stopping, then rolling back down.

Ask students:

1. What changes as the skateboarder goes up?
2. Where was the "go" when they stopped at the top?
3. Where does it come from on the way down?
4. Why do they not quite get back to their starting height?

Accept everyday language. Do not correct it yet; return to these answers at the end of the lesson.

Main teaching: Stores, transfers and systems

Introduce the eight stores with one familiar example each, then the four pathways. Use the table above.

Then introduce the system. The same event can be described differently depending on what you include:

- If the system is just a falling ball, gravity does work on it from outside.
- If the system is the ball and the Earth, the energy simply moves from the gravitational store to the kinetic store inside the system.

Students only need the idea: a system is whatever we choose to look at, and in a closed system the total energy stays the same.

Main teaching: The five changes in the specification

Work through each case. Write the stores before and after, then the pathway.

Change	Before	After	Pathway
A ball thrown upwards	Kinetic	Gravitational potential	Mechanically (gravity slows it)
A moving car hits a wall	Kinetic	Thermal (car, wall, surroundings)	Mechanically; some is carried away as sound, which ends in the thermal store of the surroundings
A trolley pushed with a constant force	Chemical (in the pusher's muscles)	Kinetic (and thermal)	Mechanically
A car braking	Kinetic	Thermal (brakes and surroundings)	Mechanically (friction)
A kettle boiling water	(Energy arrives from the mains)	Thermal (water)	Electrically to the element, then by heating to the water

Teach students to write it as a sentence:

> *Energy is transferred mechanically from the kinetic store of the car to the thermal store of the brakes.*

Activity 1: Energy transfers cartoon

Students annotate a six-frame cartoon (in the workbook) with the store that is filling and the store that is emptying in each frame, and the pathway between frames. Suggested story: a child eats breakfast, cycles up a hill, freewheels down, brakes at the bottom.

Frame	Main store emptying	Main store filling
Eating	Chemical (food)	Chemical (body)
Cycling uphill	Chemical (body)	Gravitational potential, thermal
Freewheeling down	Gravitational potential	Kinetic
Braking	Kinetic	Thermal (brakes)

Activity 2: Energy circus (optional)

Four or five stations around the room. At each, students carry out the change and write the stores before and after.

- Wind-up toy: elastic potential → kinetic
- Dropping a ball onto the bench: gravitational potential → kinetic → thermal store of the ball, the bench and the surroundings (some carried by sound)
- Rubbing hands: chemical → kinetic → thermal
- Battery and lamp: chemical → (electrically) → (by radiation and by heating) → thermal store of the surroundings
- Squeezing a stress ball and letting go: chemical (muscles) → elastic potential → kinetic and thermal

Check for understanding

1. Name three energy stores.
2. Is light a store or a transfer?
3. What is the unit of energy?
4. What happens to the energy of a car when it brakes?
5. A ball is thrown upwards. Which store empties and which fills?

6 What is a closed system?

Plenary

Return to the skateboarder. Students now rewrite the starter answers in energy language:

> *On the way up, energy is transferred from the kinetic store to the gravitational potential store. On the way down, it goes back. Some is transferred to the thermal store of the wheels and the surroundings each time, so they never get quite as high again.*

The last sentence is a preview of Lesson 8.

Possible prep

For three activities at home (for example, boiling a kettle, riding a bike downhill, bouncing a ball), write which store empties, which fills and the pathway.

Teacher notes

Common misconceptions:

1. **Energy is "used up".** It never is. It moves to a store where it is less useful, usually the thermal store of the surroundings. This is the most important correction in the unit.
2. **Light, sound and electricity are stores.** They are pathways. Insist on "transferred electrically", not "electrical energy".
3. **Energy and force are the same thing.** A force can transfer energy, but a force is a push or a pull measured in newtons; energy is measured in joules.

AQA does not test the list of stores by recall in isolation, but students must use the language accurately in every longer answer in the unit.